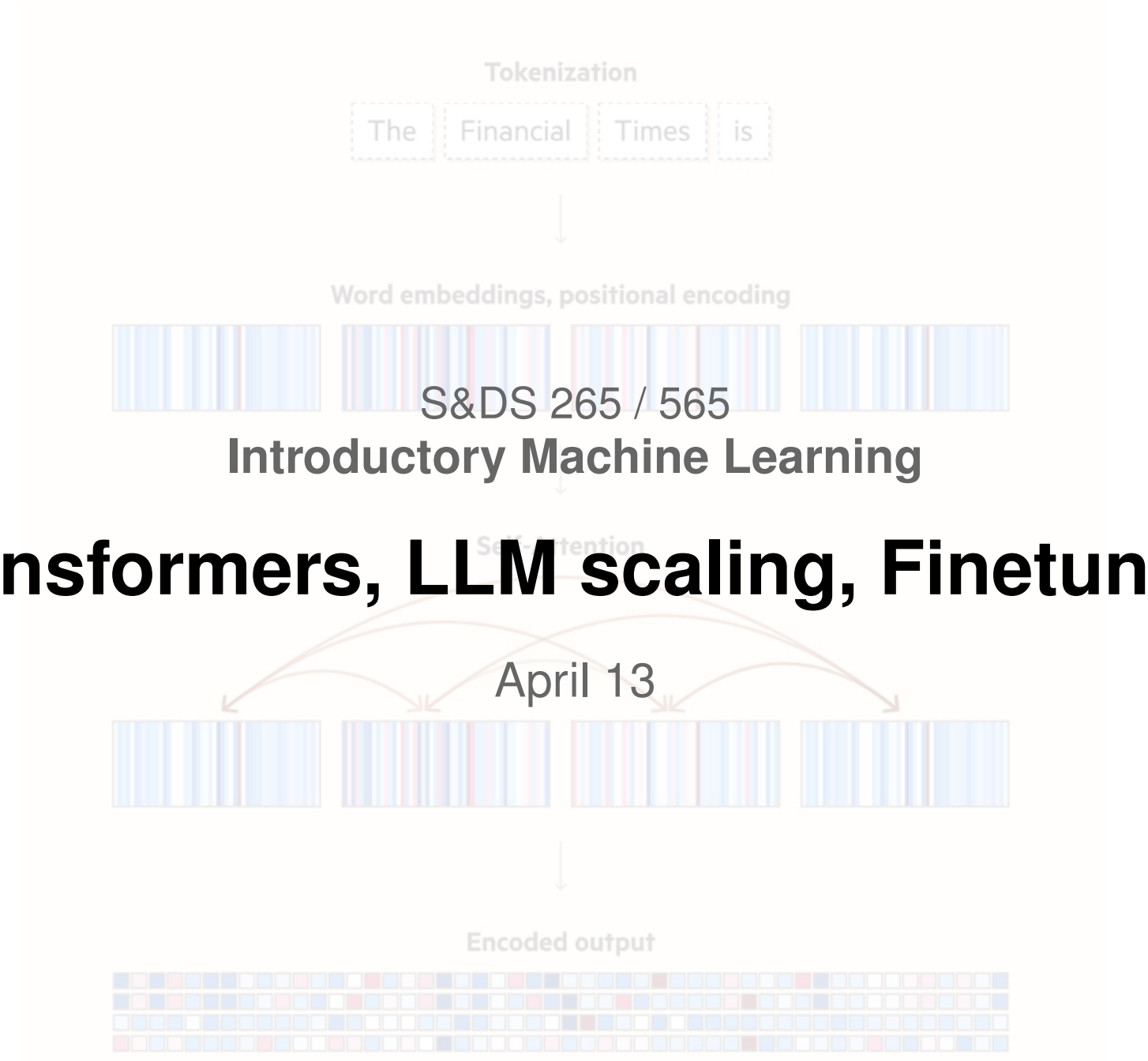


S&DS 265 / 565

Introductory Machine Learning

Transformers, LLM scaling, Finetuning

April 13

Yale

Version of 2026/04/13 at 12:42

Attention, please!

- Assignment 5 (last!) is out
- Quiz 5 (last!) this Wednesday
  - Q-learning, RL, autoencoders, LLMs
- Final exam, May 4, at 2pm
- Review sessions:
  - Yihan: Wed. April 29, 6pm
  - Dingyao: Thu. April 30, 8pm
  - Awni: Frdiay, May 1, 3pm

For today

- Scaling laws
- Finetuning

Recall: The Transformer architecture

Encoders and Decoders

Attention

- Attention is a type of alignment between related words
- Attention allows dynamic construction of word contexts, useful for predicting the next word
- Attention is computed using inner products, after mapping query and key to a common space
- Transformers process all words together, using layers of encoders and decoders, each with an attention component

Demo

Attention

This demo illustrates some of the concepts behind attention in transformers, using word embeddings. The goal is simply to show how the basic calculations underlying attention work, and how weighting embeddings with attention can change the words that are similar. As implemented in Transformers, attention can be notoriously difficult to understand.

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import networkx as nx
import gensim
```

Implementation of Attention

Embedding for current position gets transformed based on similarity to embeddings at previous positions

$$e_t \leftarrow e_t + \sum_{s<t} \alpha_{st} \left(W_V e_s \right)$$

$$= e_t + W_v \left(\sum_{s<t} \alpha_{st} e_s \right)$$

where $W_v \in \mathbb{R}^{d \times d}$ is a matrix.

Implementation of Attention

Embedding for current position gets transformed based on similarity to embeddings at previous positions

The *attention weights* are computed in terms of similarity scores:

$$w_{s,t} = (W_K e_s)^T (W_Q e_t)$$

$$\alpha_{s,t} = \frac{\exp(w_{s,t})}{\sum_{s'<t} \exp(w_{s',t})} \equiv \text{Softmax}(w_{s,t})$$

where $W_K \in \mathbb{R}^{d \times d}$ and $W_Q \in \mathbb{R}^{d \times d}$ are two matrices.

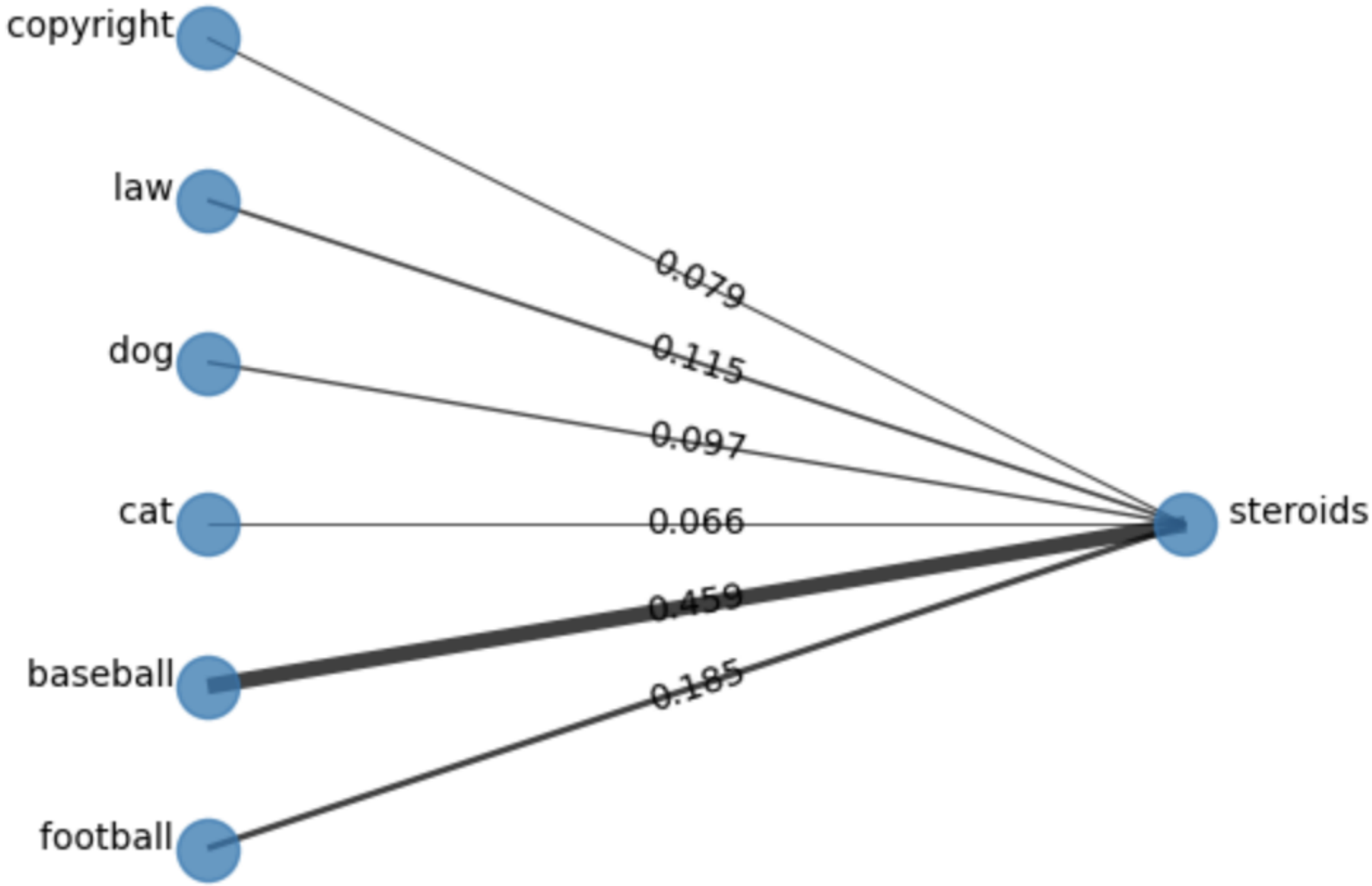
Implementation of Attention

Embedding for current position gets transformed based on similarity to embeddings at previous positions

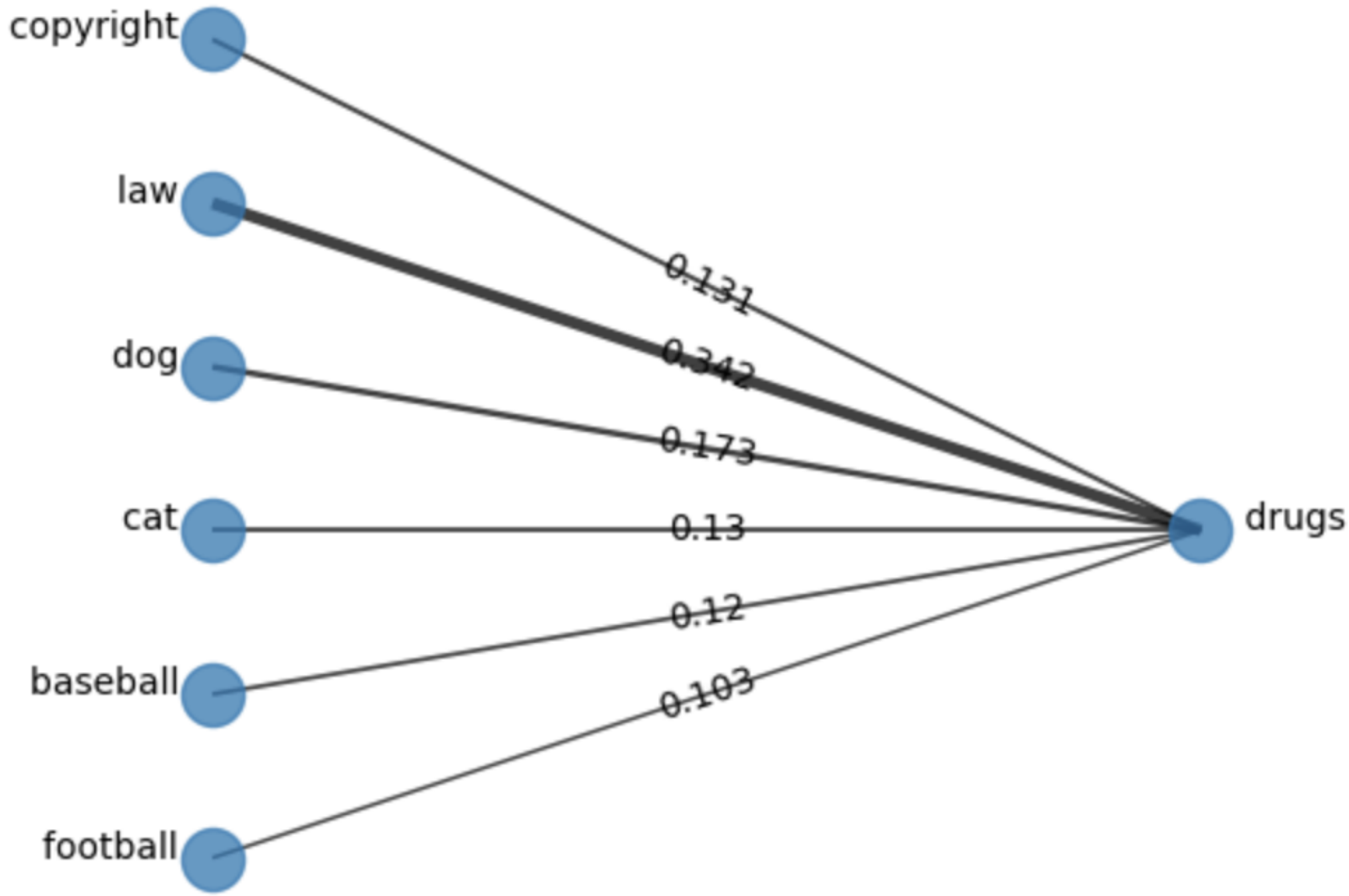
The *attention weights* α_{st} sum to one:

$$\sum_{s<t} \alpha_{st} = 1$$

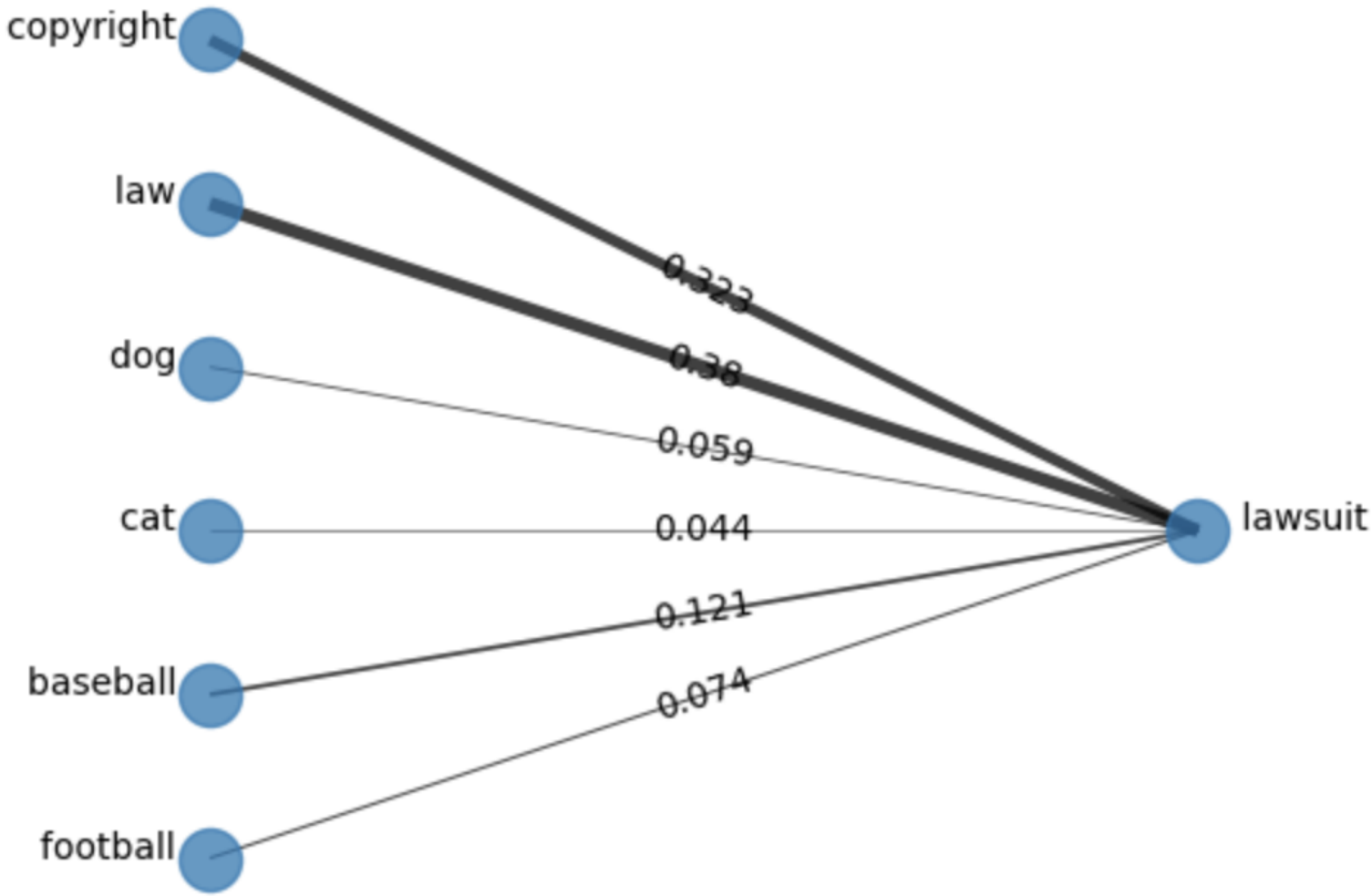
Examples from demo



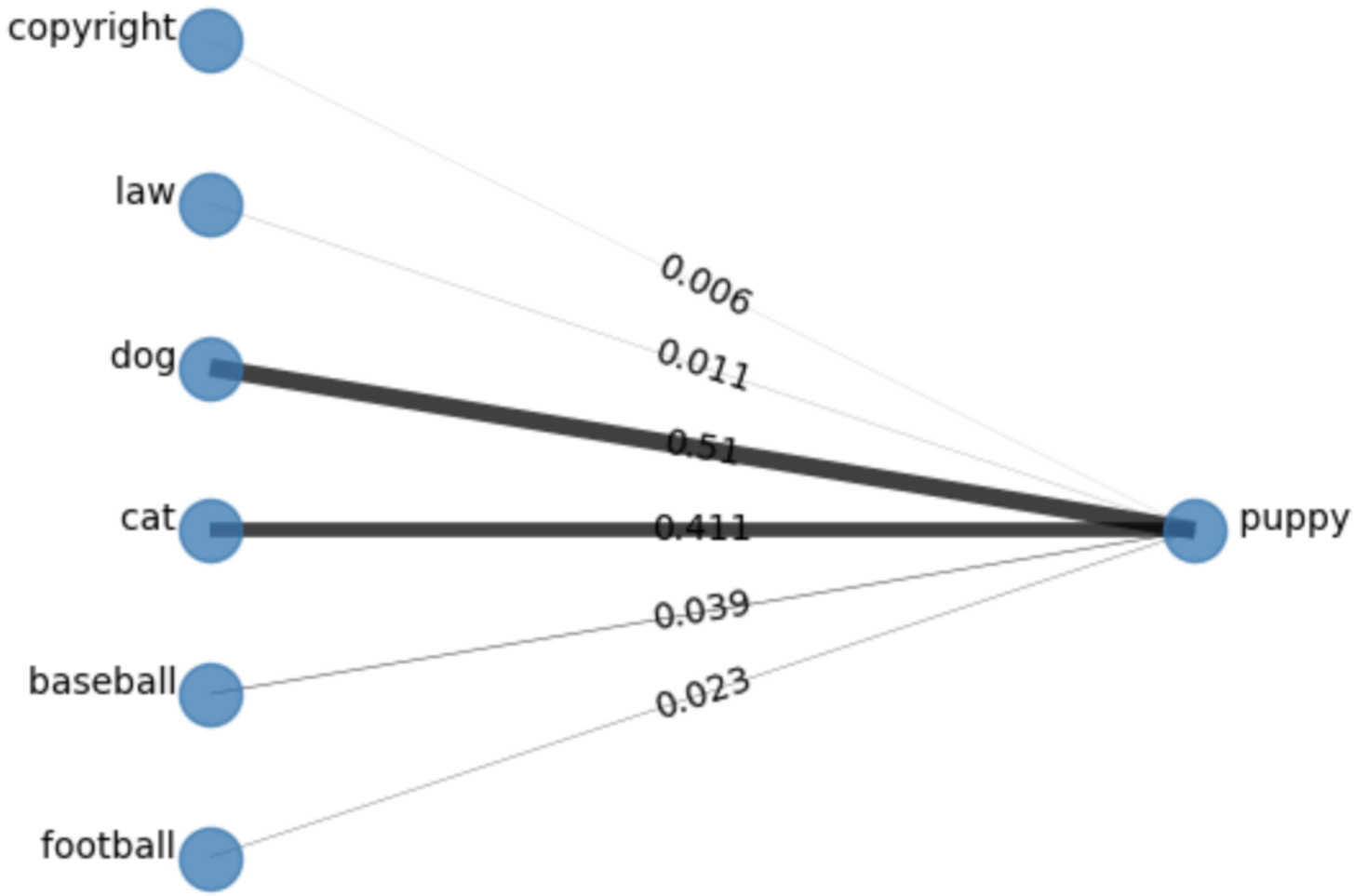
Examples from demo



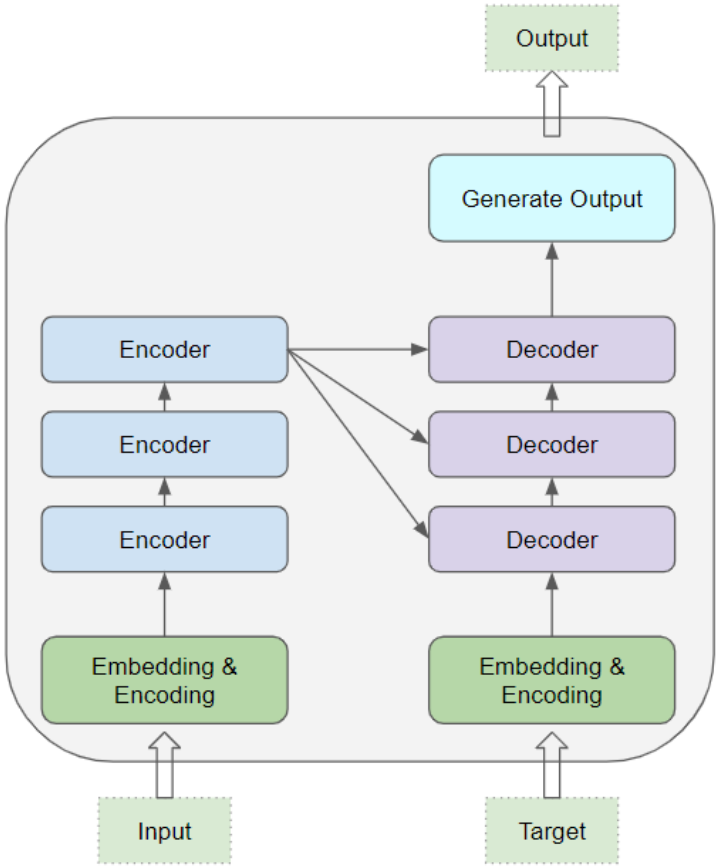
Examples from demo



Examples from demo



Transformer architecture



Transformer architecture

- Each Encoder transforms its input to generate a vector E for each source word
- Each Decoder transforms its input to generate a vector D for each target word
- The Decoder model uses *cross-attention* to attend to the Encoder vectors

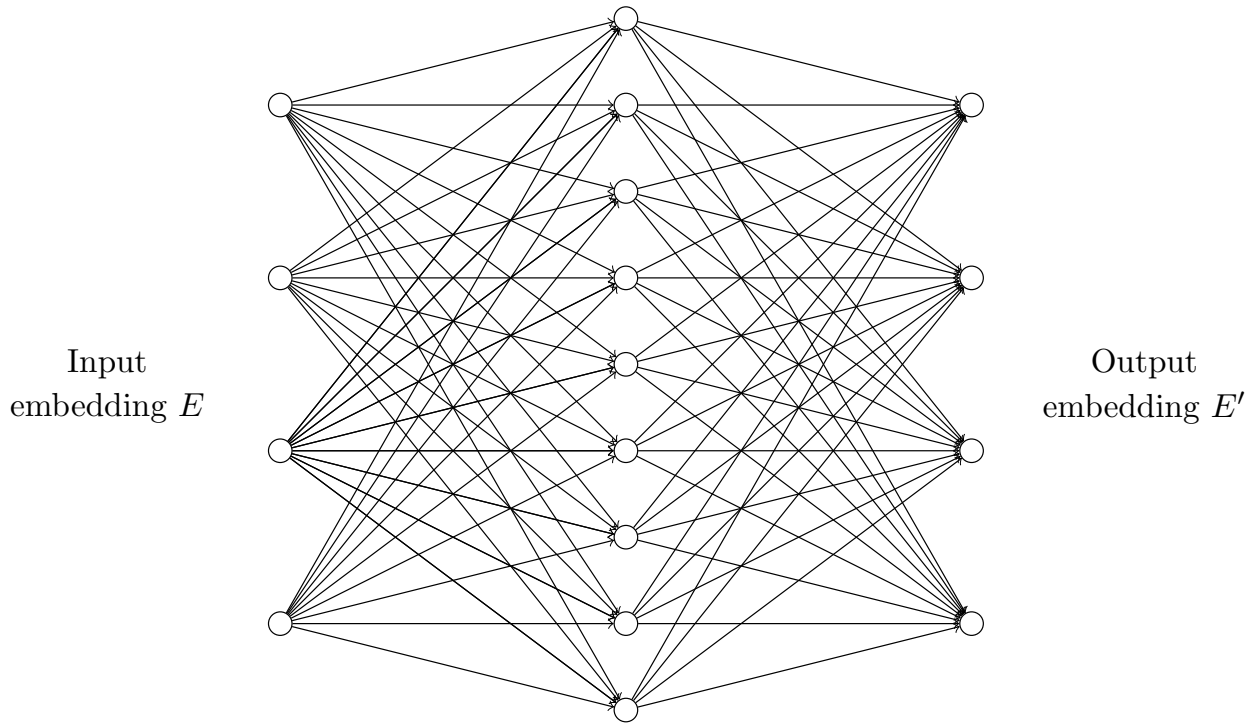
$$\text{Encoder self-attention: } Q \leftarrow E, K \leftarrow E, V \leftarrow E$$

$$\text{Decoder self-attention: } Q \leftarrow D, K \leftarrow D, V \leftarrow D$$

$$\text{Decoder cross-attention: } Q \leftarrow D, K \leftarrow E, V \leftarrow E$$

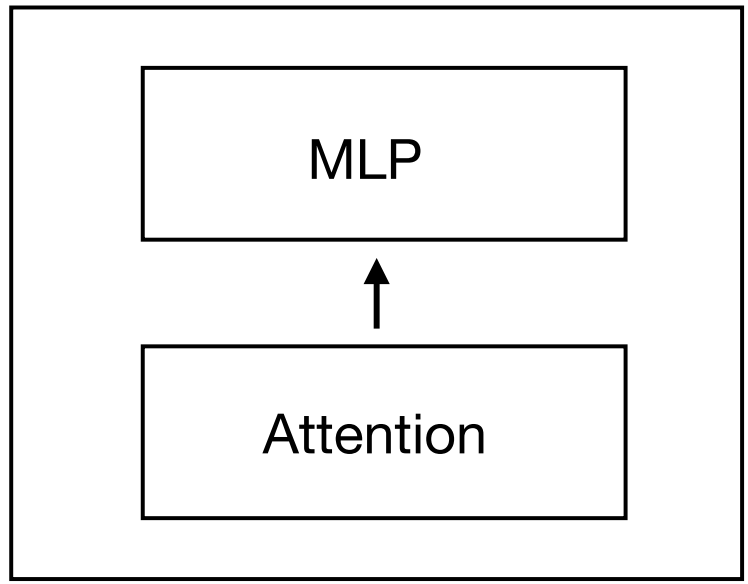
Adding nonlinearities

After attention, the mixed embeddings are transformed to vectors using an MLP:

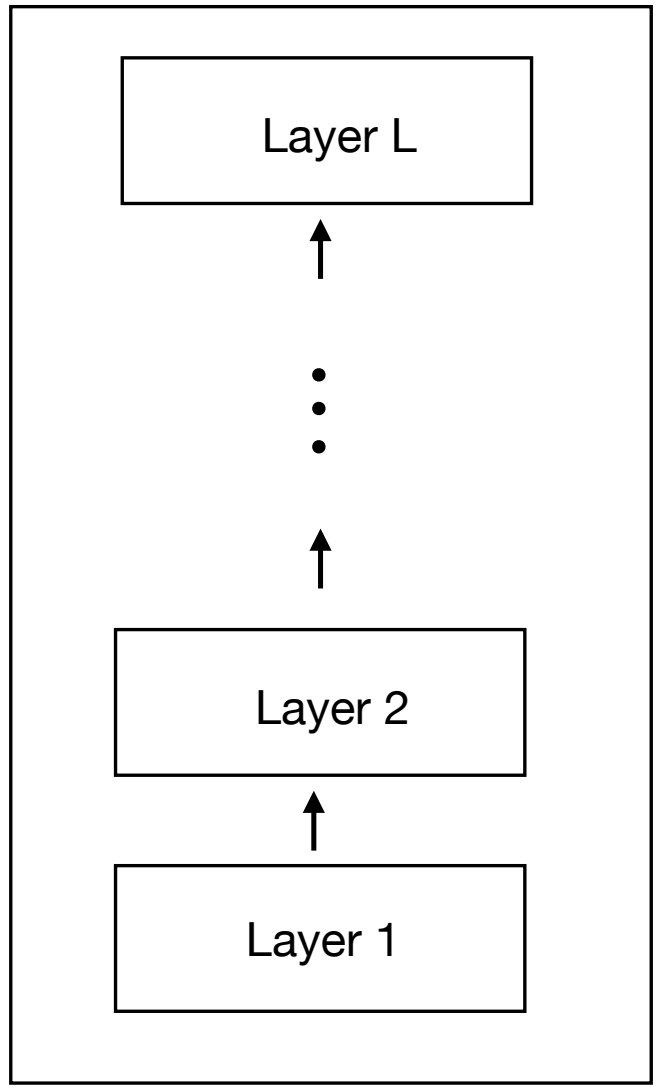


Creates new “features” to be used as embeddings in the next layer

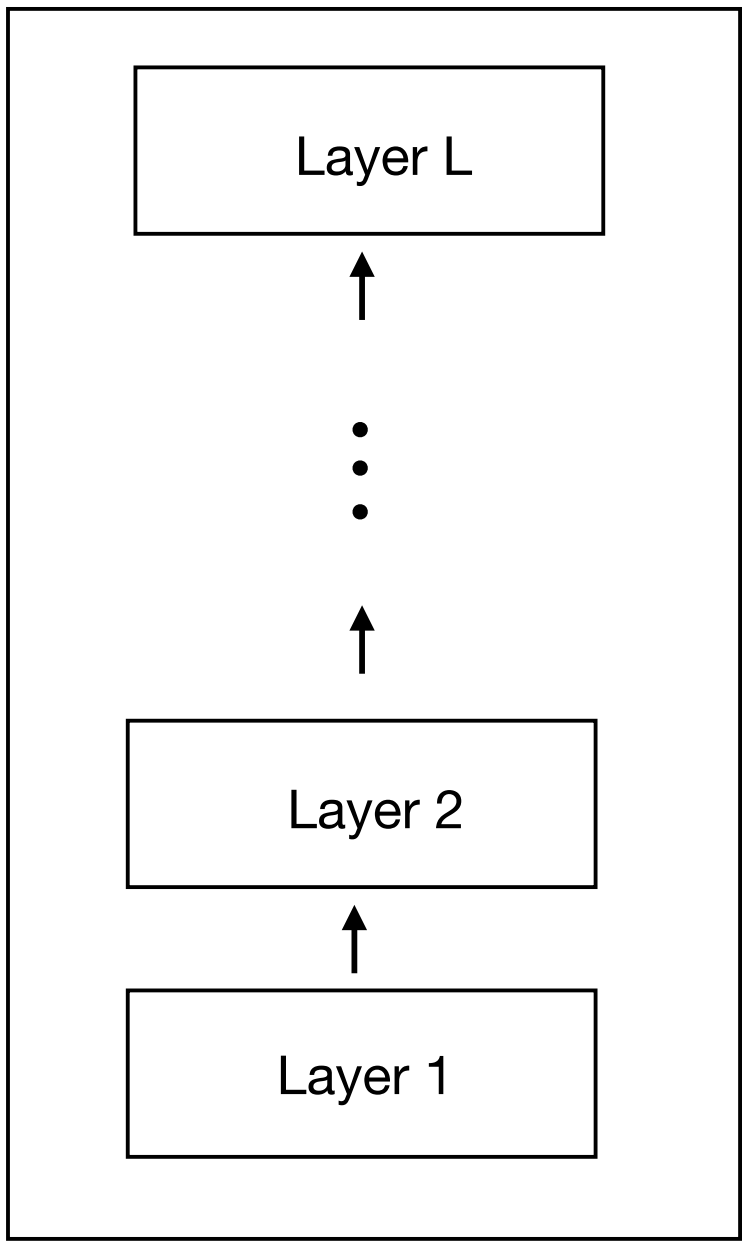
Transformer Layer

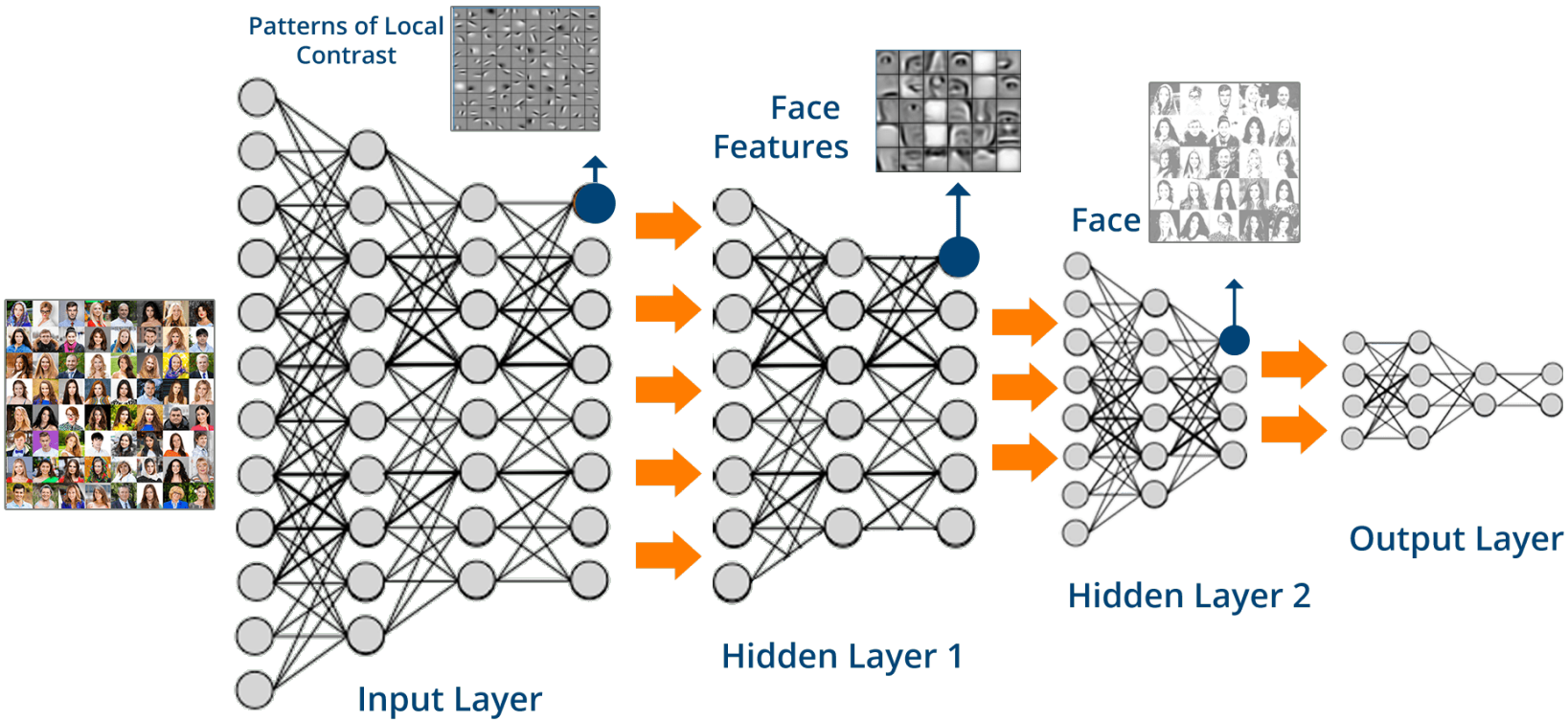


Transformer



Transformer




Patterns of Local Contrast
Face Features
Face
Input Layer
Hidden Layer 1
Hidden Layer 2
Output Layer

Recall: The Transformer architecture

How large are the models?

How does size of LLM scale?

Design choices: number of tokens V, dimension d of embeddings, and number of Transformer layers L

- Embeddings have $d \cdot V$ parameters
  - d numbers for each token, V tokens

How does size of LLM scale?

Design choices: number of tokens V, dimension d of embeddings, and number of Transformer layers L

- Self-Attention has $\approx 4d^2$ parameters
  - $d \times d$ matrix for queries (across heads)
  - $d \times d$ matrix for keys (across heads)
  - $d \times d$ matrix for values (across heads)
  - $d \times d$ matrix for combining heads

How does size of LLM scale?

Design choices: number of tokens V, dimension d of embeddings, and number of Transformer layers L

- MLP applied after has $\approx 8d^2$ parameters
  - Two layers
  - Hidden layer has $4 \cdot d$ neurons
  - Output layer has d neurons
  - $4d^2$ weights in each layer; $8d^2$ weights overall

How does size of LLM scale?

Design choices: number of tokens V, dimension d of embeddings, and number of Transformer layers L

- Embeddings have $d \cdot V$ parameters
- In each Transformer layer:
  - Attention has about $4d^2$ parameters
  - MLP has about $8d^2$ parameters
- Total trainable parameters: $12d^2 \cdot L + d \cdot V$

Size of models

GPT-2 (2019)

$$d = 1{,}600$$
$$V = 50{,}257$$
$$L = 48$$
$$12 \cdot d^2 \cdot L + d \cdot V \approx 1.5 \text{ billion parameters}$$

- Each GPT-2 Transformer layer has about 30 million parameters

Size of models

GPT-3 training data[1]:9

Dataset	# tokens	Proportion within training
Common Crawl	410 billion	60%
WebText2	19 billion	22%
Books1	12 billion	8%
Books2	55 billion	8%
Wikipedia	3 billion	3%

en.wikipedia.org/wiki/GPT-3. Note: training often refers to number of tokens *processed*; requires multiple passes through data

Size of models

GPT-3 (2020)

$$d = 12{,}288$$
$$V = 50{,}257$$
$$L = 96$$
$$12 \cdot d^2 \cdot L + d \cdot V \approx 173 \text{ billion parameters}$$

- Each GPT-3 Transformer layer has about 1.8 billion parameters

GPT-4 (2023) data and size unknown; thought to be 1.76 trillion. The human brain has about 100 trillion synapses.

Original painting by Johannes Vermeer; transformed (pixelated) by acagastya., CC0, via Wikimedia Commons

ANNALS OF ARTIFICIAL INTELLIGENCE

CHATGPT IS A BLURRY JPEG OF THE WEB

OpenAI's chatbot offers paraphrases, whereas Google offers quotes. Which do we prefer?

By Ted Chiang

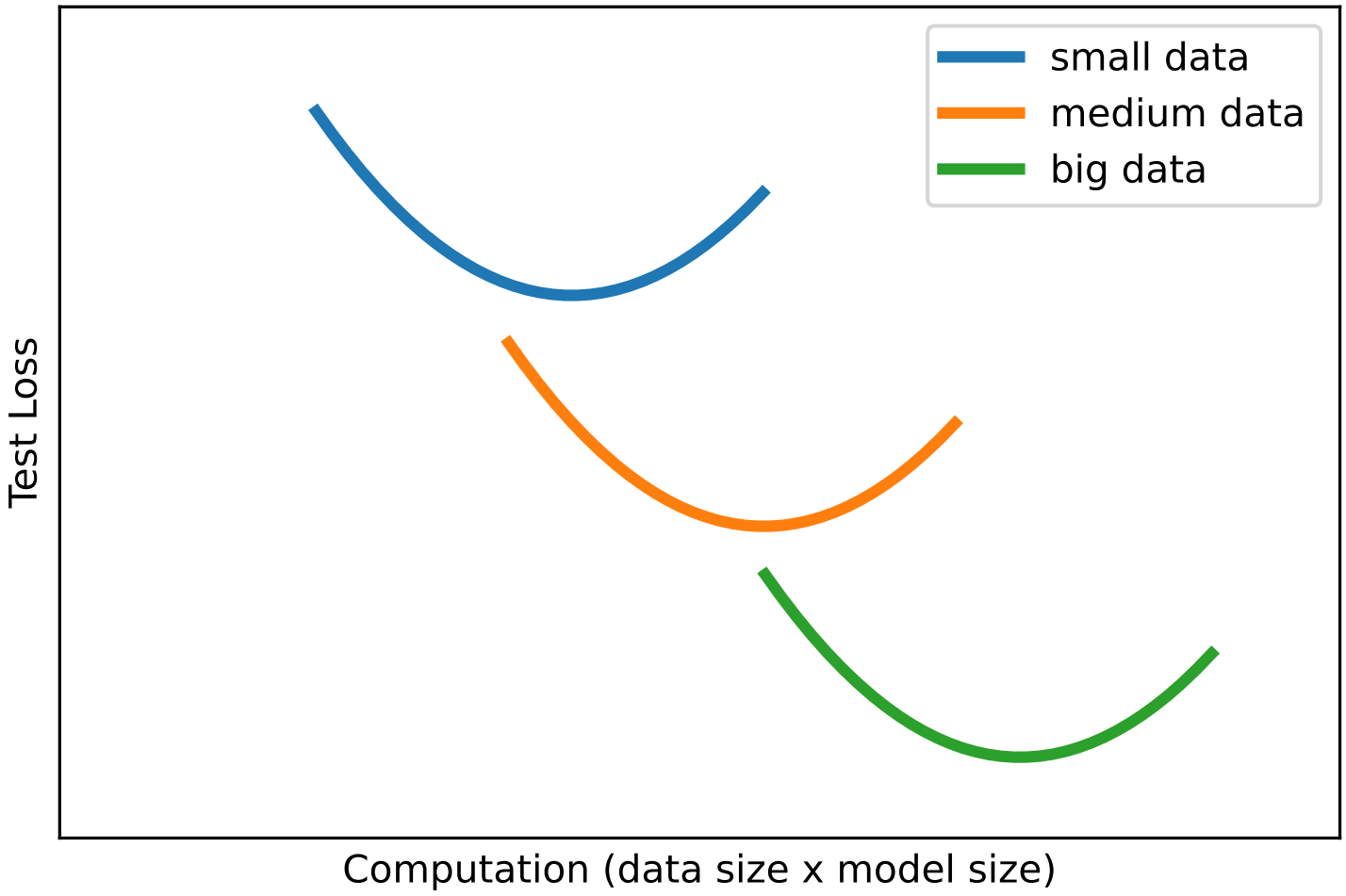
February 9, 2023

https://www.newyorker.com/tech/annals-of-technology/chatgpt-is-a-blurry-jpeg-of-the-web

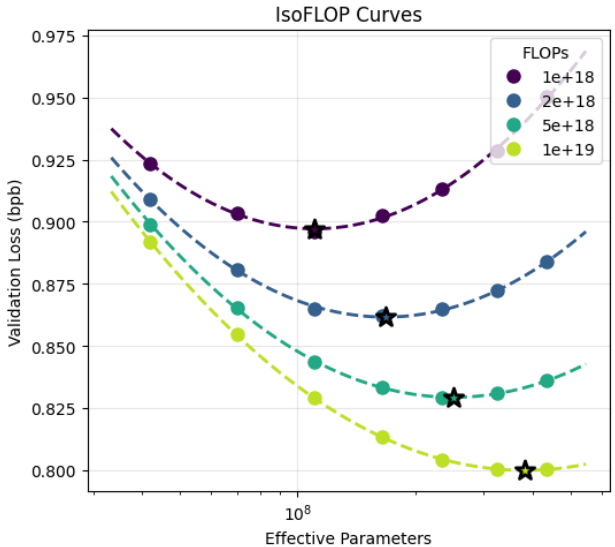
Back of the envelope calculation

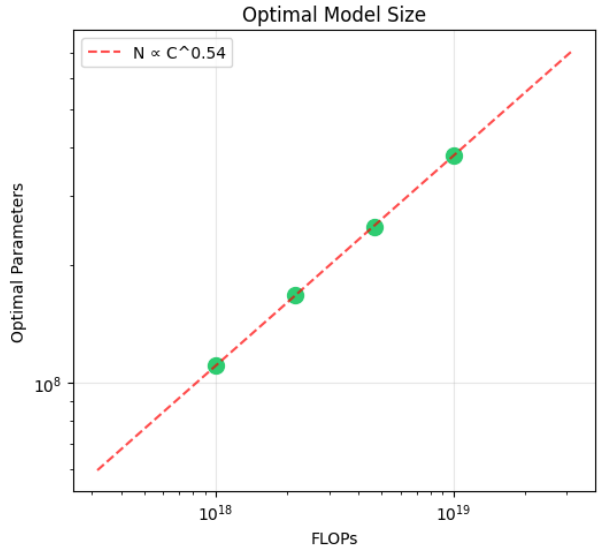
- LLM loss on test data is ≈ 1.9 nats/token
- Translates to entropy of about $1.9 / \log_e(2) \approx 2.74$ bits/token
- So, *lossless* coding of 500 billion tokens takes ≈ 171 gigabytes
- GPT-3's 175 billion params uses 346 gigabytes (16-bit precision)
- GPT-3 could, in principle, memorize most of the internet!

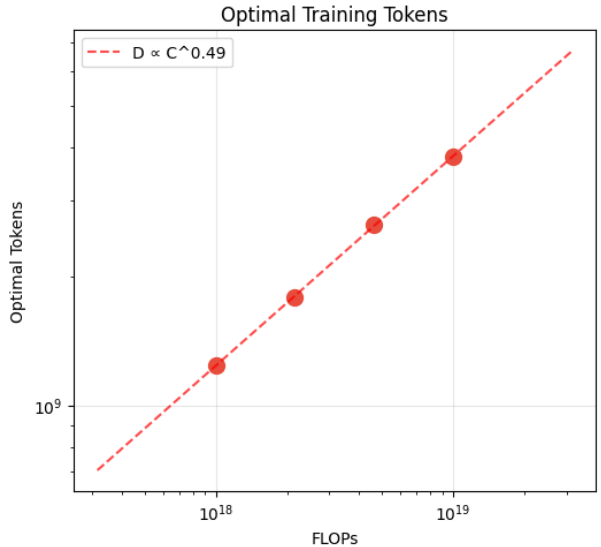
Scaling behavior of LLMs: Performance



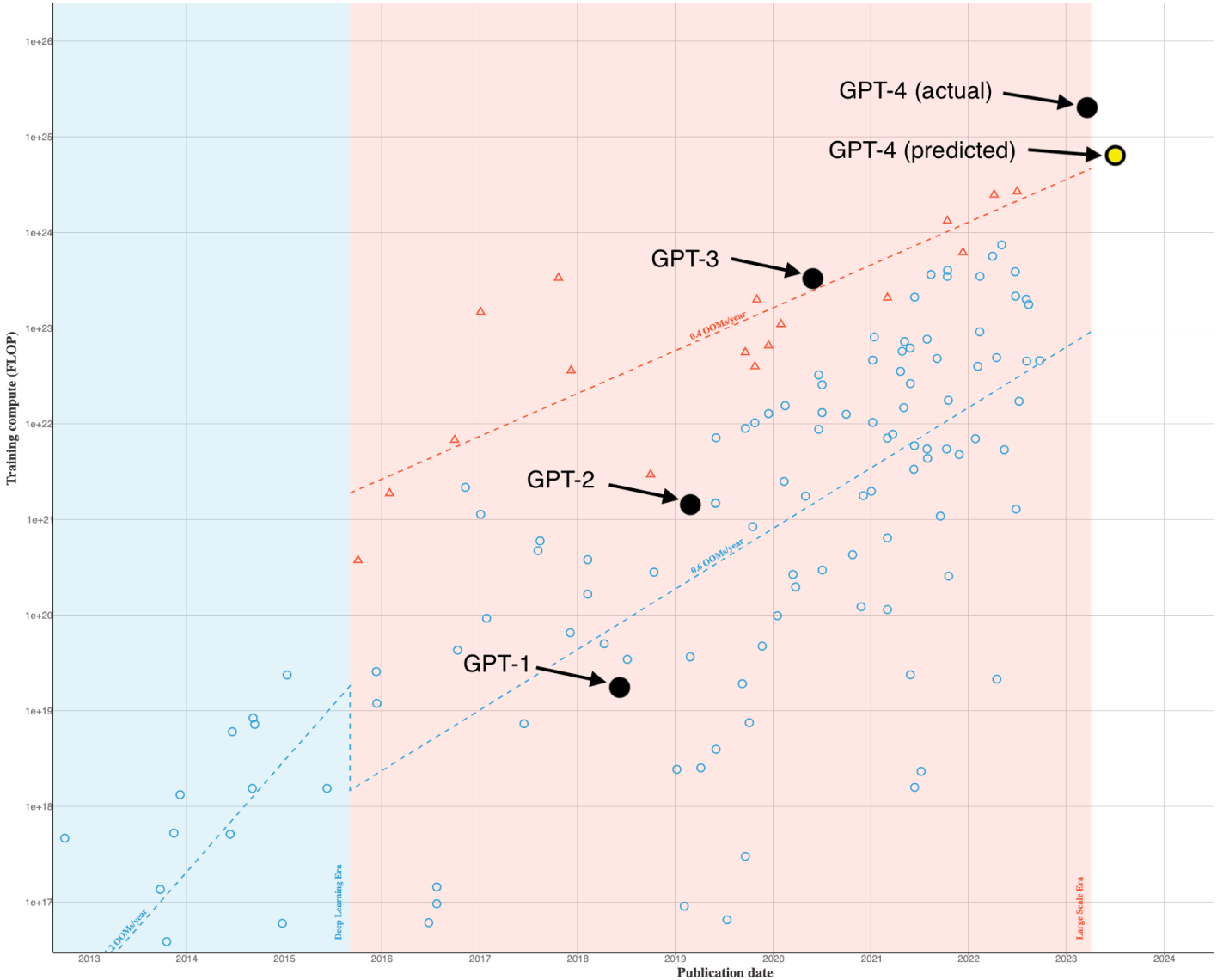
Scaling behavior of LLMs: Performance







https://github.com/karpathy/nanochat


GPT-4 (actual)
GPT-4 (predicted)
GPT-3
GPT-2
GPT-1
0.4 OOMs/year
0.6 OOMs/year
2 OOMs/year
Deep Learning Era
Large Scale Era
Training compute (FLOP)
Publication date
1e+26
1e+25
1e+24
1e+23
1e+22
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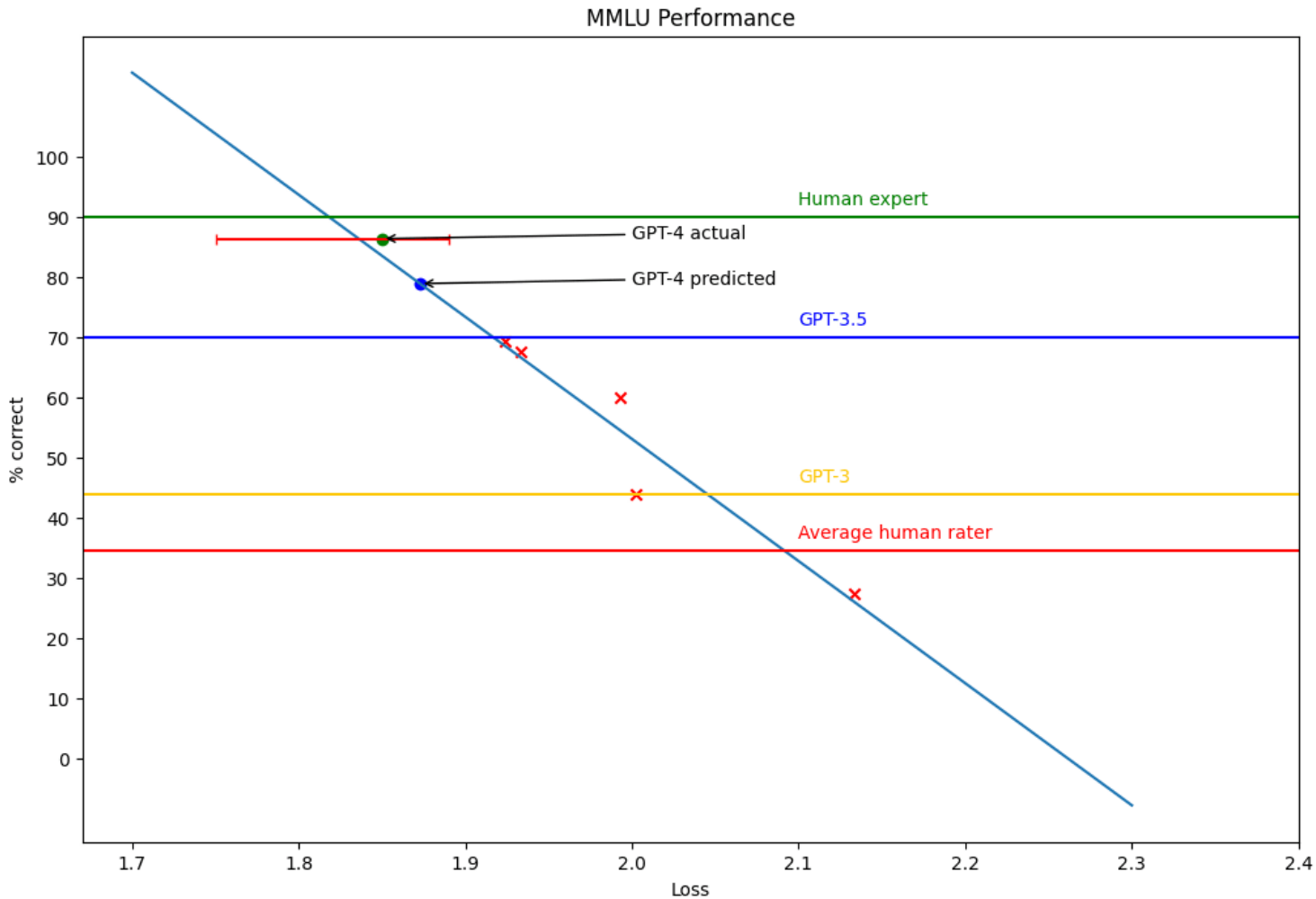


MMLU: Massive Multitask Language Understanding, https://en.wikipedia.org/wiki/MMLU, https://paperswithcode.com/dataset/mmlu

Sutton's "Bitter Lesson" (2019)

"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin."

---

http://www.incompleteideas.net/IncIdeas/BitterLesson.html

Generating text

The generation process

Rolling a D50,257

- Transformer assigns a score to each token
- Converted to a probability
- Generating next token equivalent to rolling weighted 50,257-sided die

Remixing Noon

Text generated from Channel Skin by Jeff Noon

"The whole forest had been anesthetised, her temples wired, her senses stimulated, her eyes were not on Eva, not on Eva, not on Eva, not on anybody in that same realm, the land of dreams and nightmares."

Viability: 0.000000326%

https://chatbotslife.com/notes-on-remixing-noon-generative-text-and-markov-chains-84ff4ec23937

Generation

Generation algorithm has two “knobs” to control:

1. *Top-k*: Restricts to just the top k tokens
   - Low k: Low variability
   - High k: Higher variability

Generation

Generation algorithm has two “knobs” to control:

2. *Temperature T*: Lower temperature favors more likely tokens
   - Low temperature: Low variability
   - High temperature: High variability

Generation

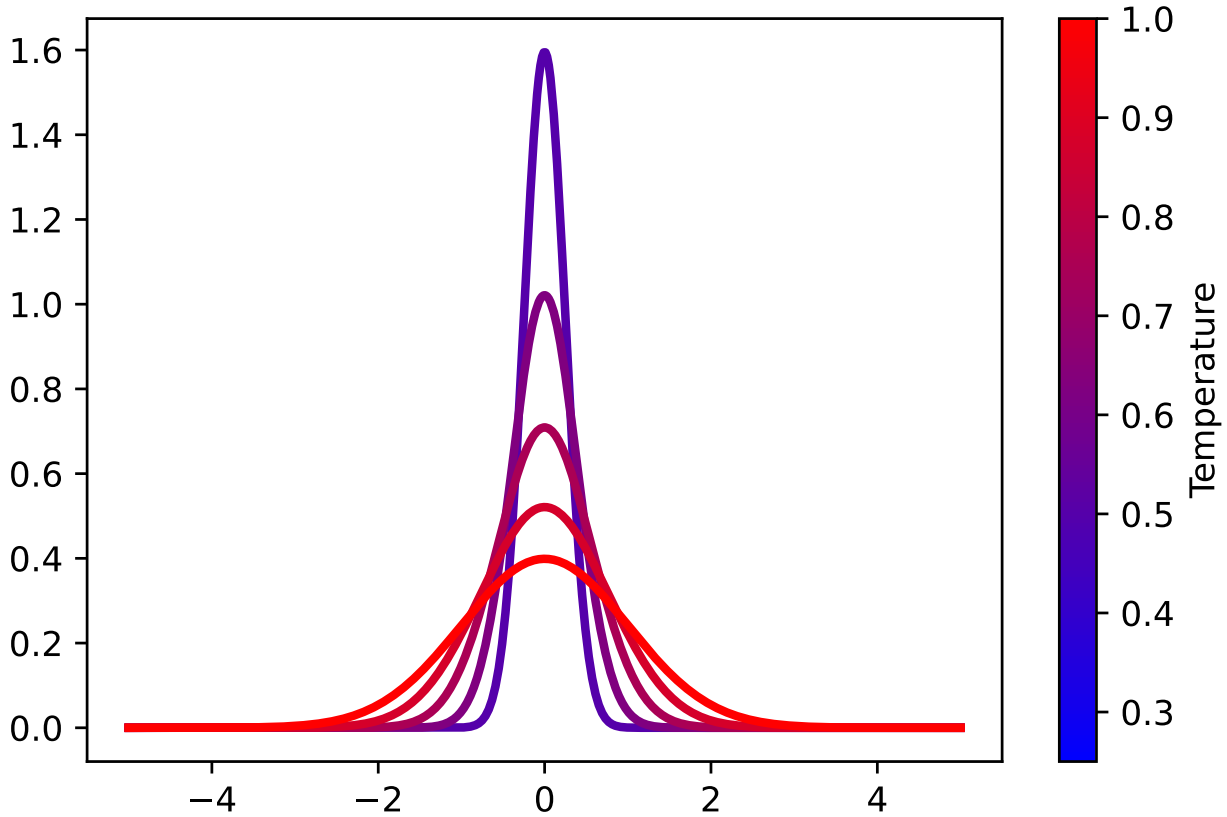
At temperature T, if the scores of the top k tokens are

$$\text{logit}_1, \text{logit}_2, \ldots, \text{logit}_k$$

the weights on the die are proportional to

$$e^{\text{logit}_1/T}, e^{\text{logit}_2/T}, \ldots, e^{\text{logit}_k/T}$$

Temperature



Temperature $T = 1$

One day, Timmy met a monster. The monster was big and green, but it was harmless. Timmy was scared, but he wanted to be brave. He walked closer and closer to the monster. Suddenly, the monster snapped its fingers! Timmy was so scared that he started to cry. The monster said, “Don’t be scared! I’m harmless!” But Timmy was still sc

One day, Timmy met a monster. The monster had big teeth and green skin. Timmy was scared and ran away. But the monster called out, “Wait! I don’t want to hurt you. I just want to be your friend.” Timmy stopped and looked back. “Really?” he asked. “Yes, I permit you to be my friend,” the monster replied. Timmy was surprised. He didn’t expect

---

https://github.com/karpathy/llama2.c

Temperature $T = 0$

One day, Timmy met a monster. The monster was big and scary. Timmy was scared and ran away. The monster chased Timmy. Timmy ran and ran until he found a big tree. He climbed up the tree and the monster couldn't reach him. Timmy was safe. Timmy saw a bird and said, "Hello bird, can you help me? The monster is chasing me." The bird said, "

One day, Timmy met a monster. The monster was big and scary. Timmy was scared and ran away. The monster chased Timmy. Timmy ran and ran until he found a big tree. He climbed up the tree and the monster couldn't reach him. Timmy was safe. Timmy saw a bird and said, "Hello bird, can you help me? The monster is chasing me." The bird said, "

Temperature $T = 1.5$

One day, Timmy met a monster. The monster was big and scary. Timmy didn't know what to do. He felt bitter because he wanted to play with the monster. The monster said, "Hello, Timmy. Can you show me around the forest?" Timmy wasn't sure if he believed him or not. He said, "I'm sorry, do you have room for crawlers?" The monster guesses sad

One day, Timmy met a monster. The monster was big and scary, but Timmy wasn't scared. "Hi," said Timmy. The monster was surprised. He couldn't understand what Benny had said. "What are you up to?" he asked. Timmy picked him up one by one and he saw doziles of Ell Party coming in on a streak street. "I amdri catientased!"

Complete this sentence: The quick brown fox

"The quick brown fox jumps over the lazy dog."

(A classic — it's a pangram containing every letter of the alphabet.)

Now complete it at a slightly higher temperature

"The quick brown fox sprinted through the amber dusk, chasing something it would never quite catch."

Again

"The quick brown fox, inexplicably wearing a monocle, demanded to speak with the manager of the forest."

Now complete it at temperature T=2.0

"The quick brown fox xylophone underneath several Tuesday moons, dripping with forgotten algebra."

(Illustrating the point — high temperature completions become incoherent as the probability distribution over tokens flattens out and unlikely/random tokens get sampled more frequently.)

Making the models useable

Finetuning

Recall: Machine learning frameworks

- Supervised
- Unsupervised, self-supervised
- Reinforcement learning
- Representation learning

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- Supervised
- Unsupervised, self-supervised
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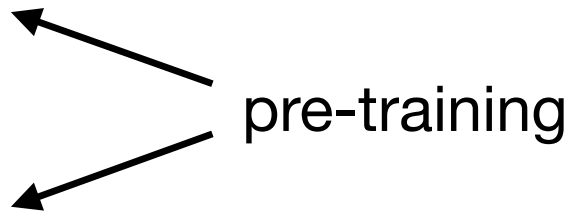

Recall: Machine learning frameworks

- Supervised
- Unsupervised, self-supervised
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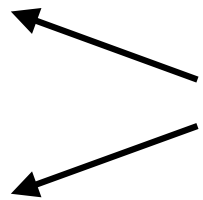


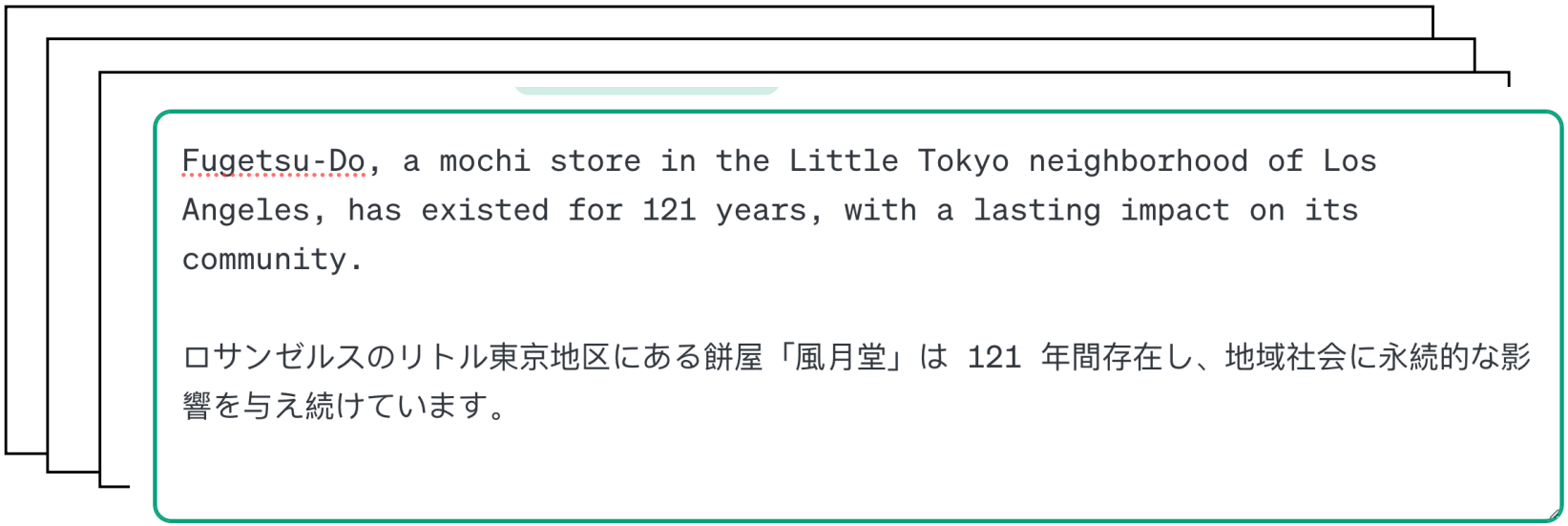
making LLMs useful

Recall: Supervised learning

Goal: Accurately assign label $y \in \{\texttt{dog}, \texttt{cat}\}$ to an image x.

"Learn" from a large database of examples $\{(x_i, y_i)\}$.

Recall: Supervised learning



Goal: Accurately translate input $x_1, \ldots, x_m$ to output $y_1, \ldots, y_m$.

"Learn" from a large database of examples $\{(x, y)\}$.

This is called *sequence-to-sequence learning* (seq2seq)

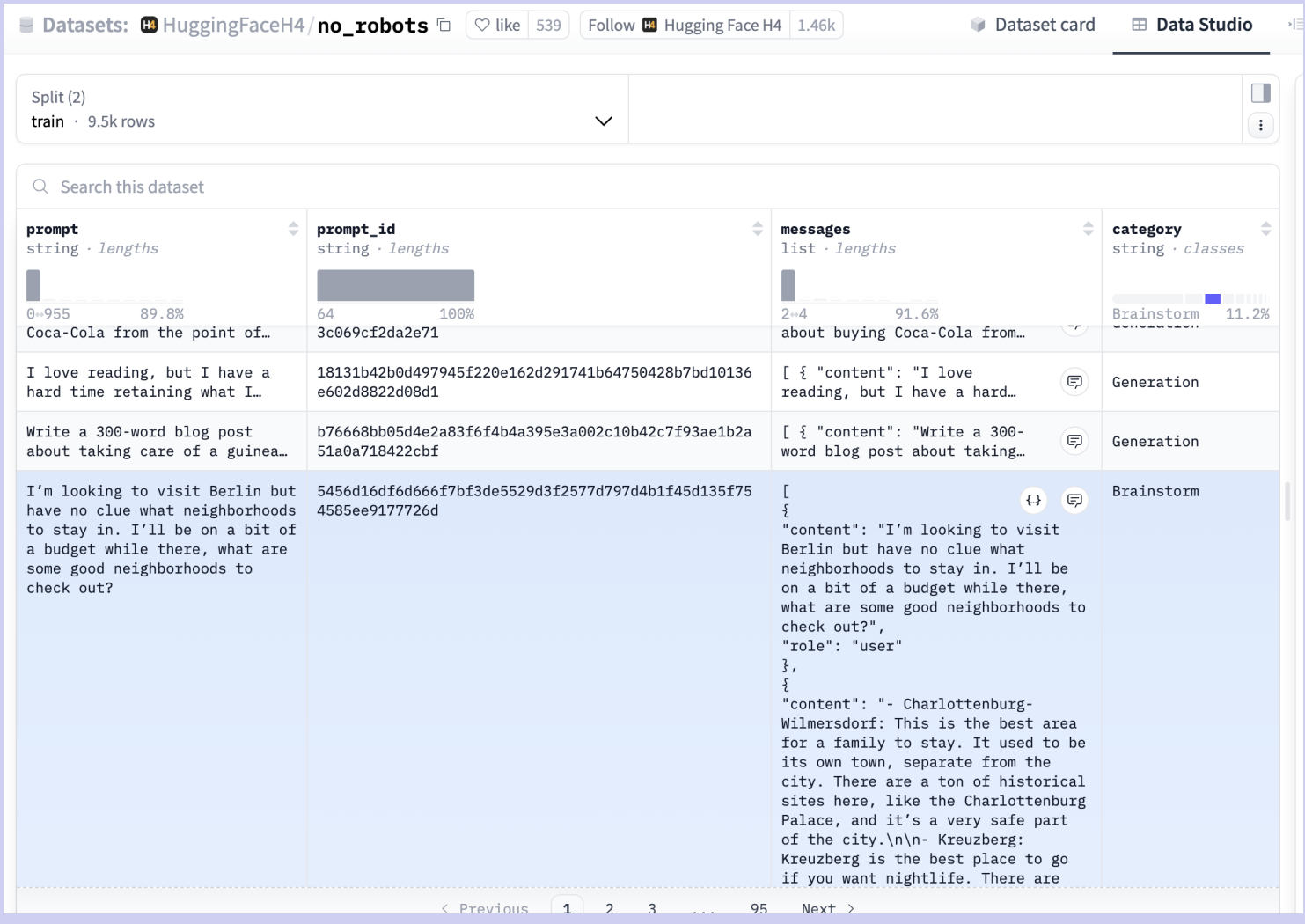
Finetuning

Finetuning takes a pre-trained LLM, and feeds it a database of examples of how it should “translate” input to output.

Examples are here:
`huggingface.co/datasets/HuggingFaceH4/no_robots`

Categories: Generation, Open QA, Brainstorm, Chat, Rewrite, Summarize, Coding, Classify, Closed QA, Extract

Example training data



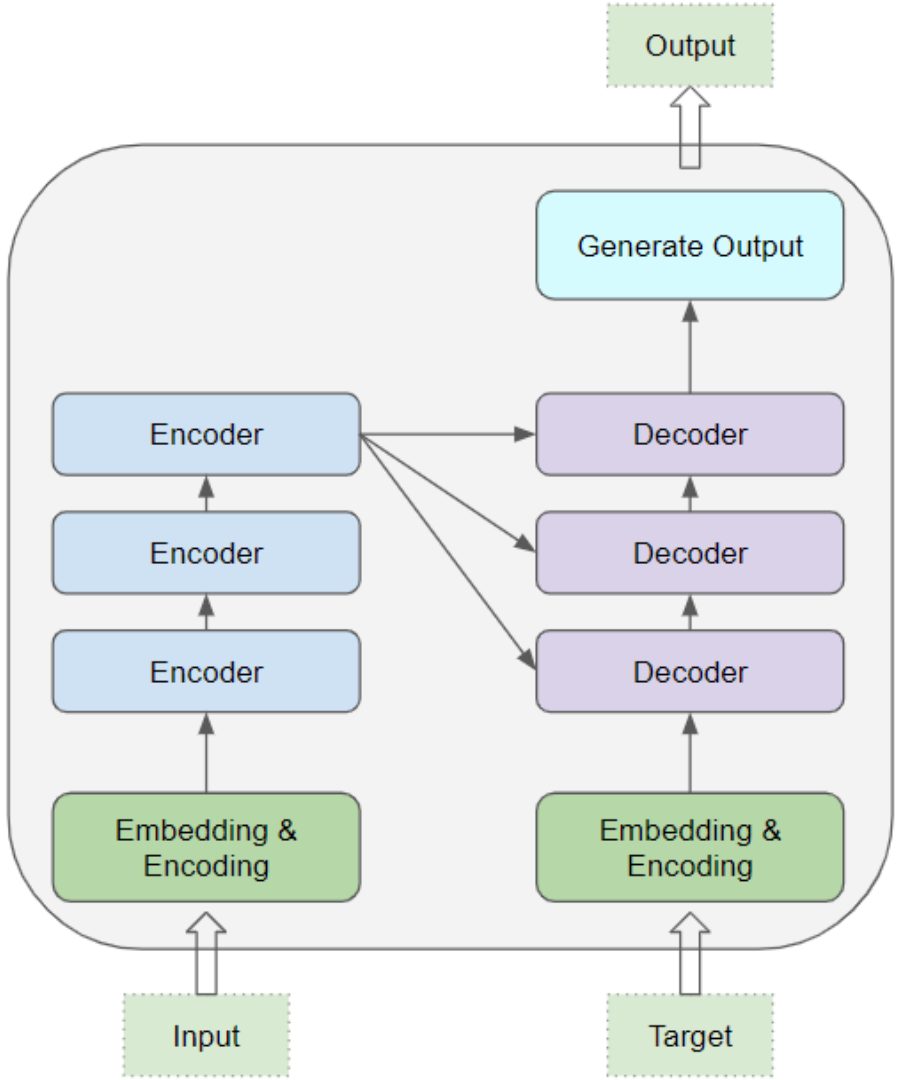
huggingface.co/datasets/HuggingFaceH4/no_robots

Training process

1. Prompt is read in and embedding / encoding vectors are processed just as in generation.
2. Target sequence is generated (predicted) token-by-token
3. Parameters are iteratively updated to make the target sequences more and more likely

After training is complete, updated LLM has been "finetuned" to the tasks in the training data

Transformer finetuning architecture



Finetuning fineprint

- The LLM is huge (175 billion parameters for GPT-3)
- Finetuning training data are too small to adjust all the parameters
- Special techniques are used to make “small” changes to the model (e.g. LoRa)

Making the models useful

Reinforcement learning

Reinforcement learning

- An agent interacts with an environment
- The actions the agent takes change the state of the environment
- The agent receives rewards for each action, and seeks to maximize the total cumulative reward

Reinforcement learning is a framework for sequential decision making to achieve a long-term goal.

Reinforcement learning: Motivating examples

- Learning to walk or ride a bike
- A robot vacuum cleaning up the house
- Playing chess, backgammon, Atari games, etc.
- Drug discovery, personalized health, energy management

Characteristics

- RL is inherently sequential
- In between supervised and unsupervised learning
- Agent can't act too greedily; needs to be strategic

The aim of RL is to learn to make optimal decisions from experience

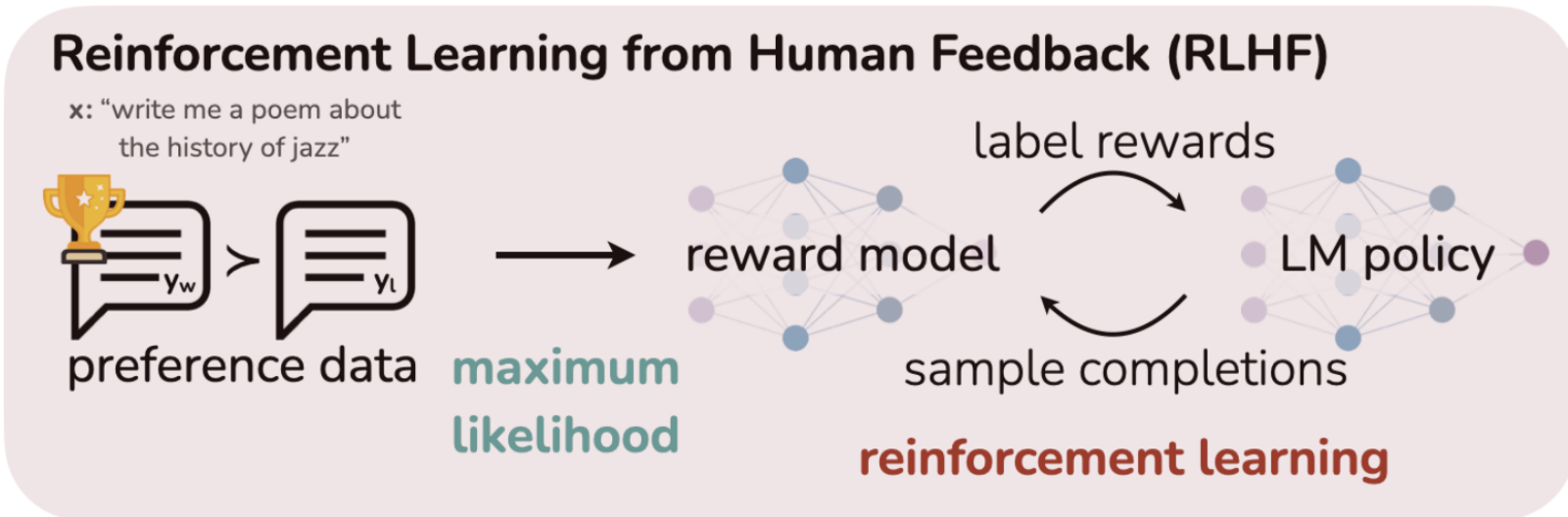
RLHF

RLHF — Reinforcement Learning from Human Feedback

- Key to RL is the *reward signal*
- First step of RLHF: Learn reward function from human preferences
- Given reward, learn policy to improve LLM outputs

RLHF

RLHF — Reinforcement Learning from Human Feedback



arxiv.org/abs/2305.18290

Policy optimization

In Reinforcement Learning, a *policy* is way of choosing actions to optimize long-term rewards

- Here, the policy governs which tokens to generate
- The rewards are the desirable characteristics of text as learned from human judges
- The policy is iteratively improved to generate text that is
  - *Helpful*, *Correct*, *Coherent*, *Appropriately Simple/Complex*
- These characteristics change with the domain

See, e.g., https://huggingface.co/datasets/nvidia/HelpSteer

Summary

- Transformer architecture
- Scaling laws: Diminishing returns?
- Generating new text
- Making the models useful:
  - Finetuning: Training on human-generated responses
  - RLHF: Learn to predict human judgements of quality (reward)
  - RLHF: Use estimated rewards to improve quality (policy)